Research Paper 01 Summary

The New VIIRS 375 m Active Fire Detection Algorithm

Problem Statement: AI-Based Detection and Classification of Industrial Fires (PS26162)

Original Paper: Schroeder et al. (2014), *Remote Sensing of Environment*

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1 Basic Information

- **Paper Title:** The New VIIRS 375 m Active Fire Detection Data Product: Algorithm Description and Initial Assessment
- **Authors:** Wilfrid Schroeder, Patricia Oliva, Louis Giglio, and Ivan A. Csiszar (2014)
- **Sensor Covered:** VIIRS (Visible Infrared Imaging Radiometer Suite) 375-meter resolution sensor on NASA/NOAA satellites.

2 What is this Paper About? (In Simple Words)

This research paper explains how NASA's primary satellite sensor (VIIRS) detects fires on Earth from space.

- **How the satellite sees heat:** The satellite has special infrared cameras. It compares the temperature of a small ground area with the temperature of the land surrounding it.
- **How it flags a fire:** If a ground point is significantly hotter than its surrounding neighborhood, the computer algorithm automatically marks that point as an active fire.
- **Why 375 meters matters:** Older satellites looked at 1 km $\times$ 1 km patches of land at a time. VIIRS looks at smaller 375 meter $\times$ 375 meter squares. This makes it 3 to 4 times better at catching small, localized heat spots.

3 The Big Problem Revealed in the Paper (Section 4)

The most important takeaway for our project is in **Section 4** of the paper:

> *The satellite algorithm only checks if an area is hot. It does not know WHAT is on the ground.*

Because of this limitation:

- Factories, oil refineries, gas flares, and power plants generate high heat every single day.
- NASA FIRMS marks all of these industrial chimneys and flare stacks as "fires".
- There is no built-in way to distinguish between an actual dangerous forest fire and a normal working factory.

4 How This Helps Our Project (PS26162)

4.1 1. Scientific Proof for Our Problem Statement

When presenting the project or answering questions, this paper gives us official proof from NASA's own scientists that raw satellite fire data (FIRMS) cannot separate industrial facilities from wildland fires. Our AI system exists to fix this exact blind spot.

4.2 2. Deciding the Map Search Radius on OpenStreetMap (OSM)

The paper shows that one satellite pixel covers an area of roughly **375 meters by 375 meters** (and up to 800 meters near the edge of the satellite's view).

- **Practical rule for our map:** When our system receives a thermal alert coordinate from NASA, it searches within a **400 to 500 meter radius** on OpenStreetMap to check if an industrial factory, power plant, or refinery is located there.

4.3 3. Simple Numbers (Features) to Feed into Our AI Model

The paper defines the exact numbers provided by the satellite for every detected spot:

- **Spot Temperature (T_4):** How hot the target spot is.
- **Background Temperature (T_5):** How warm the surrounding land is.
- **Temperature Difference ($\Delta T = T_4 - T_5$):** How much hotter the spot is compared to its background.
- **Fire Radiative Power (FRP):** The total heat energy emitted in megawatts (MW).

Industrial flare stacks burn at very high temperatures ($> 1000\,\mathrm{K}$) in a small concentrated spot, giving them distinct temperature differences compared to wide, low-temperature forest fires.

4.4 4. Building the Daily Data Ingestion Pipeline

This paper serves as the reference guide for downloading and reading daily alerts from NASA FIRMS, including the confidence score (*low, nominal, high*) and location coordinates.